DIGITAL DESIGN AND COMPUTER ARCHITECTURE · DDCA

Design and Simulation of a Four-Bit Binary Arithmetic Calculator Using Combinational Logic

Team : Batch 1

PROJECT SCOPE

Integrates concepts from DDCA Modules 1–6

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01 · SYSTEM OVERVIEW

A four-bit calculator turns binary inputs into arithmetic results

01

Binary inputs

Processes two four-bit binary inputs through a combinational arithmetic circuit.

02

Arithmetic circuit

Represents binary numbers, uses logic gates, and generates arithmetic results.

03

Binary output

Outputs are verified in Logisim Evolution, including carry generation.

Relevance: the project links digital system design to computer architecture.

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Full adders are the logic foundation of the calculator

FOUR-BIT FOUNDATION

Module 1: four-bit unsigned values range from 0000 to 1111. Logic gates, truth tables, Boolean expressions, and two's complement establish the digital foundation.

Module 2: combinational outputs depend on current inputs. Half adders, full adders, and ripple-carry adders build arithmetic circuits.

Subtraction, multiplexers, comparators, and a multi-operation ALU are proposed extensions, not current features.



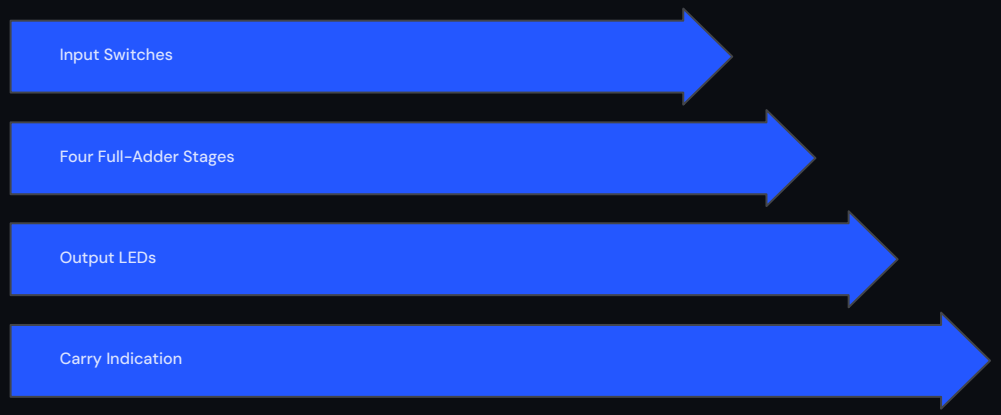
FULL-ADDER EQUATIONS

$$\text{Sum} = A \text{ XOR } B \text{ XOR } C_{in}$$

$$\text{Carry} = (A \text{ AND } B) \text{ OR } (C_{in} \text{ AND } (A \text{ XOR } B))$$

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Logisim connects four one-bit adders into a four-bit circuit



Illustrative circuit diagram · not a Logisim Evolution screenshot

Four one-bit stages pass each carry to the next to form a ripple-carry adder.

MODULE CONNECTIONS

Arithmetic circuit as an ALU building block; flip-flops and registers add storage between clock cycles.

CPU registers hold operands and results; RAM and ROM provide data and instructions.

A RISC-V datapath connects register file, ALU, memory, program counter, and control unit; larger operations and processor integration are future extensions.

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04 · VERIFICATION

Expected results verify the carry path

FULL-ADDER TRUTH TABLE

A B Cin	Sum Cout
0 0 0	0 0
0 0 1	1 0
0 1 0	1 0
0 1 1	0 1
1 0 0	1 0
1 0 1	0 1
1 1 0	0 1
1 1 1	1 1

EXPECTED FOUR-BIT ADDITION TEST CASES

A + B	Decimal check	Expected output	Carry
0011 + 0010	3 + 2 = 5	0101	0
0111 + 0001	7 + 1 = 8	1000	0
1111 + 0001	15 + 1 = 16	0000	1; 1 0000 overall

PLACEHOLDER · CASE 1

Input/output case 1

PLACEHOLDER · CASE 2

Input/output case 2

PLACEHOLDER · CARRY

Carry-out case

Change input switches and observe output LEDs in Logisim Evolution. Compare expected binary results with circuit ou
individual input bits, carry signals, and output bits. Label visuals as expected results when actual screenshots ar

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From logic gates to a processor datapath

Logic gates

Arithmetic circuit

ALU

Processor

Implemented now: binary representation, Boolean logic, full-adder stages, a four-bit combinational arithmetic circuit, and simulation-oriented verification. **Module 1:** bits and logic gates. **Module 2:** combinational logic and ALU foundations. **Modules 3–5:** sequential logic, registers and memory, and the RISC-V datapath provide architecture context; they are not standalone calculator features.

Applications: ALUs, digital calculators, processor datapaths, embedded systems and IoT devices, and digital electronics education. **Module 6:** future extension toward more operations, new instructions, memory-mapped outputs, and processor integration. **Future improvements:** subtraction, operation selector, seven-segment display, operand registers, simplified RISC-V integration, pipelining, caching, and embedded applications.

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THANK YOU

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